higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 2

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This marking guideline consists of 5 pages.

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QUESTION 1: BASIC CONCEPTS AND ROAD-BUILDING TERMINOLOGY

1.1 The formation level is the designed ground level where final construction will take place.

1.2 Asphalt is a thick black substance used for making the surface of a road.

1.3 A road sign shows what is happening on the road.

1.4 The verge is an outside space of the road usually covered in grass to extend the road in future.

1.5 The road shoulder is the space outside the road where vehicles and people stop.

(5 × 2) **[10]**

QUESTION 2: ROAD SURVEYING

2.1 A line level is an instrument used to measure levels over a distance connected by a cable or line. (3)

2.2
- Water level
- Spirit level
- Dumpy level (Any 2 × 1) (2)

2.3
- Stretch the line between two points and secure it.
- Mark out the desired level from the starting points.
- Place the level on the line in the middle.
- Check the bubble on the level to see if it is at the centre of the glass level-measuring tube.
- It the bubble is not at the centre of the glass tube, adjust the line until the bubble is level or centre. (5)

2.4 BC – beginning of curve
EC – end of curve (2)

2.5
- From 1, measure back along the distance IT 1 and IT 2.
- Hammer in pegs at these points and mark the exact point T1 and T2 with nails.
- Hook a steel tape over each nail and mark the centre at O which is the point where the tape intersect when reading R.
- Swing the radius at any point on the tape over the peg O. (4)

2.6
- Set up the tripod by extending its legs.
- Secure the tripod by pressing its legs firmly into the ground with your feet.
- Adjust the length of the legs so that the top of the tripod is at a comfortable height.
- Extend the length of the tripod to above mid chest or shoulder height, roughly forming a triangle. (4)

[20]

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QUESTION 3: SOIL GRAVEL AND ROAD-BUILDING MATERIALS

3.1 The moisture content of soil is the percentage✓ of water in the soil that help the soil particles stick together during compaction✓, while optimum moisture content is the phrase used when the soil particles are not floating in excess water✓ or too dry to compact.✓ (2 + 2) (4)

3.2
- Take a small wet-soil sample and roll it between the hands and on a glass plate to form a thread or sausage shape.
- Roll the thread gradually until its thick is reduced to a diameter of ±3 mm.
- Stop rolling the soil at this point and dry out the sample until the thread begins to crumble.
- Weigh the sample.
- Dry the soil sample in an oven (preheated between 105 °C and 110 °C) and record its weight once it is dry.
- Lastly calculate the moisture content of the sample. (6)

3.3
- Standard cone-penetrometer test
- Three tin containers with dimensions 40 mm deep and 55 mm diameter
- Stopwatch (3)

3.4
- It helps the soil particles to stick together during compaction.
- It makes soil compaction easy. (2)

3.5
- Too much water on the soil
- Air in the soil (2)

3.6
- Cracks on the road
- Disrupted storm-water drainage system
- Rise and fall on the soil (3)

[20]

QUESTION 4: ROAD DRAINAGE

4.1
- To keep the road surface free from excessive water and debris
- To prevent erosion (2)

4.2
- At the toe of a cut where the side slope is stable
- Across the road on the downhill side of a cut
- At the toe of a fill (3)

4.3 4.3.1 To collect water through it from one side to the other

4.3.2 To remove water from a drain next to the toe of a road-reserve boundary

4.3.3 To determine the degree of damage expected and prevent damage to embankment

(3 × 2) (6)

4.4
- Potholes
- Sometimes the whole road can wash away (2)

4.5
- Natural water table
- Irrigation canals and dams
- Rainfall infiltration (3)

4.6
- Surface drainage✓ is used to collect water running on the road surface away from the road.✓
- Subsurface drainage✓ is used to collect water seepage getting into the road layers away from the road.✓ (4)

[20]

QUESTION 5: EARTHWORKS

5.1 5.1.1
- Backacter for digging trenches
- Dragline for bulk excavation

5.1.2
- Bulldozer for ripping and loosening rock earth in preparation for excavation
- Front-end loader for loading sand and soil into the truck

5.1.3
- Road/Tandem roller for compacting soil and road materials
- Pneumatic tired roller for compacting a wide range of soil of low to moderate plasticity such as salt and clay

(3 × 4) (12)

5.2
- Use the correct personal protective equipment.
- Avoid incorrect use of the plant.
- Clear the site prior to any major earthwork. (3)

5.3 5.3.1
- Keep the side tidy and get rid of rubbish.
- Provide handrails for stairs and gangways over trenches.

(Any ONE)

5.3.2 Provide helmets to workers.

5.3.3
- Ensure that materials are properly stacked and not likely to be blown.
- Materials should not be thrown down haphazardly from a height.

(Any ONE)
(3 × 1) (3)

5.4 5.4.1
- The location of underground utility hazards must be determined before starting an excavation.
- Avoid the operation of equipment near the top edge of an excavation because of the chance of slope failure.

5.4.2
- Provide formal safety programmes for all new employees.
- Give periodic refresher training for each worker.

5.4.3
- Shut down the engine and do not allow smoking during refuelling.
- Do not allow workers to ride on equipment unless proper seating is provided.

(3 × 2) (6)

5.5 Earthwork includes all work on the ground like excavation, hauling and compaction. (1)

[25]

QUESTION 6: LAYER WORKS

- Perform a soil-classification test to verify the type of soil that must be stabilised.
- Using the percentage of fine soil, determine the type of stabiliser to add to the soil.
- Hand and spread the material onto the road.
- Mix the stabiliser with the soil.
- Level and compact the layer.
- Allow the soil layer to set and harden. (Any 5 × 1) **[5]**

TOTAL: 100